

Project: Mini-Project

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Course:

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Sciences and Bioengineering Sciences

GPU Computing

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1 Intro

Throughout all parts of the mini-project, it was chosen to compare two **operations** namely: addition & multiplication. To start, a comparison between the operations will be done, as requested in part 1. Following that, the kernel will be updated to execute on multiple elements simultaneous (one-to-many) mapping.

For the third part, the compute intensity will be increased by applying a loop count factor over the computations.

The final part will compare the local vs global memory location, as indicated in the scheme¹ in the assignment.

2 Structure

TODO

3 Methodology

For each iteration of the combination of values to be measures, **20** runs where measured. The first 5 runs where discarded to allow the system to stabilize. This is in accordance to the recommendations in [1]. For each measured value, if applicable a Cov range chart is produced and will be reference, these can be found in §9. The acceptable range of the Cov is taken from [2].

¹http://parallel.vub.ac.be/education/gpu/labs/miniproject_step4_with_local%20memory.jpg

4 Part 1: Addition vs Multiplication

4.1 Setup

TODO

4.2 Memory Bandwidth

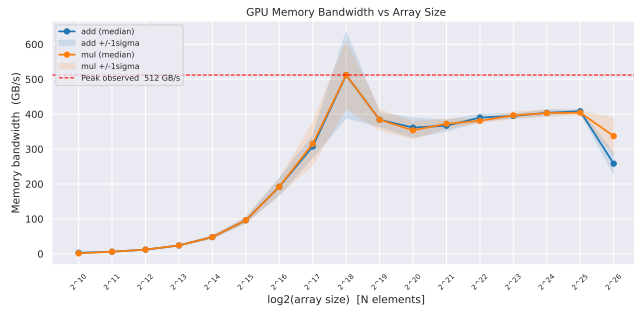


Figure 1:

TODO

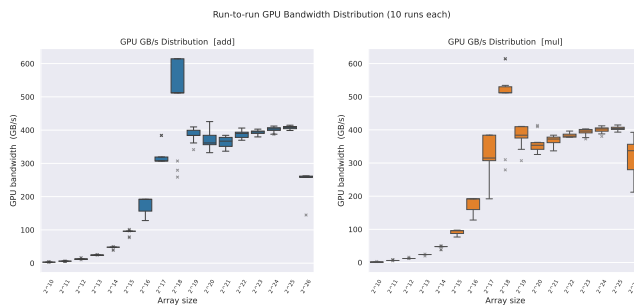


Figure 2:

4.3 Compute Throughput

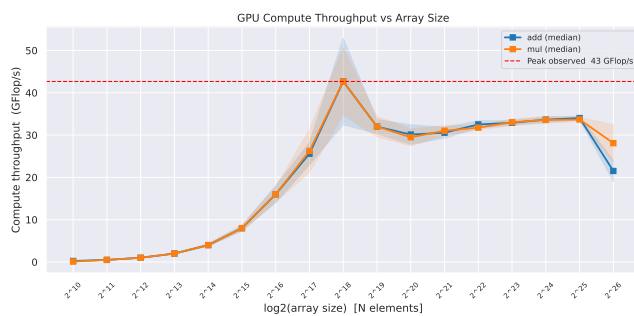


Figure 3:

5 Part 2: Elements Per Thread

5.1 Setup

The array size is fixed for all benchmark variations performed in this part to 2^{22} , based on the data gathered in part 1 and the workgroup size is set to 64.

5.2 Memory Bandwidth

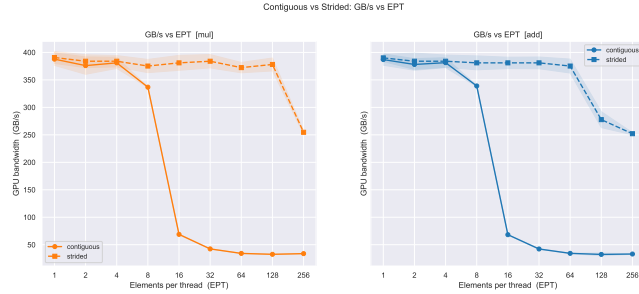


Figure 4:

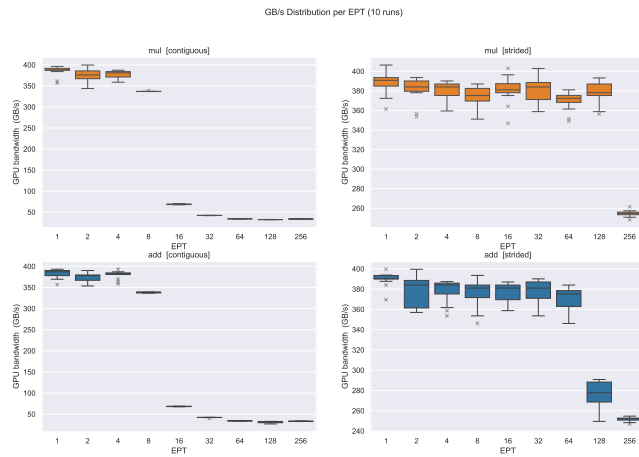


Figure 5:

5.3 Time

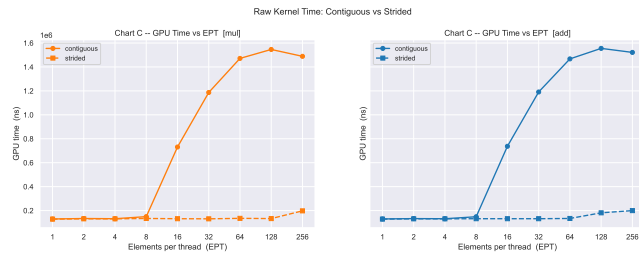


Figure 6:

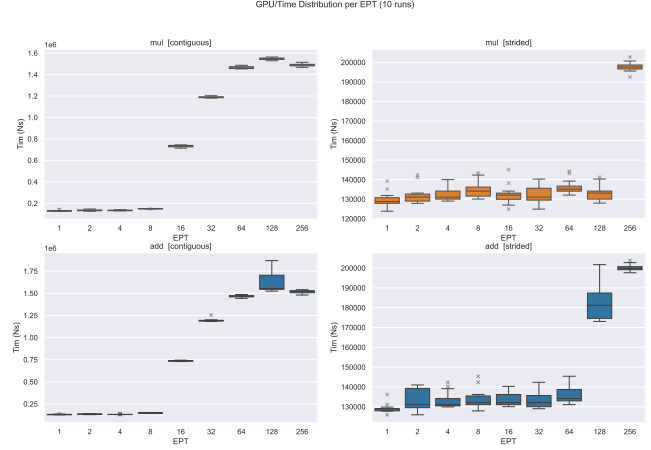


Figure 7:

6 Part 3: Roofline Model

6.1 Setup

The array size is fixed for all benchmark variations performed in this part to 2^{22} , based on the data gathered in part 1 and the workgroup size is set to 64.

6.2 Model

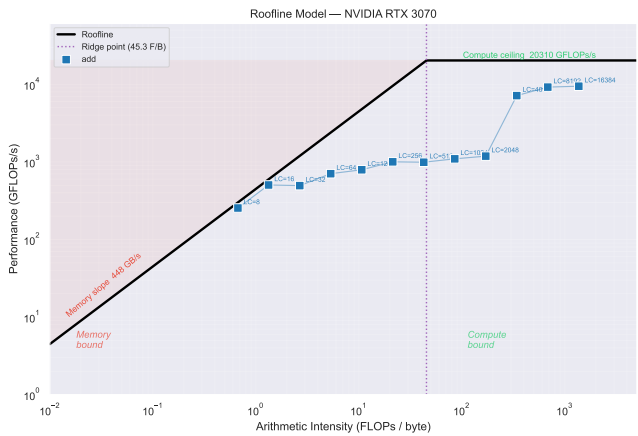


Figure 8:

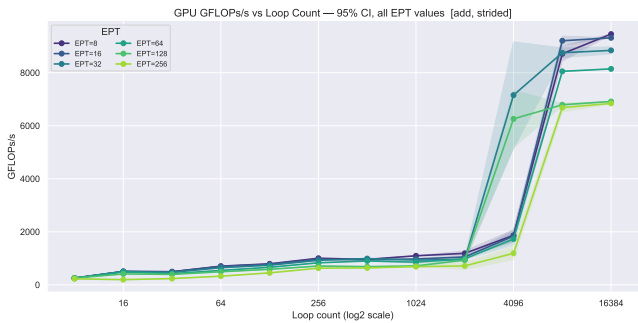


Figure 9:

7 Part 4: Local vs Global

8 Extra: Desktop vs Macbook

9 Appendix

9.1 Part 1

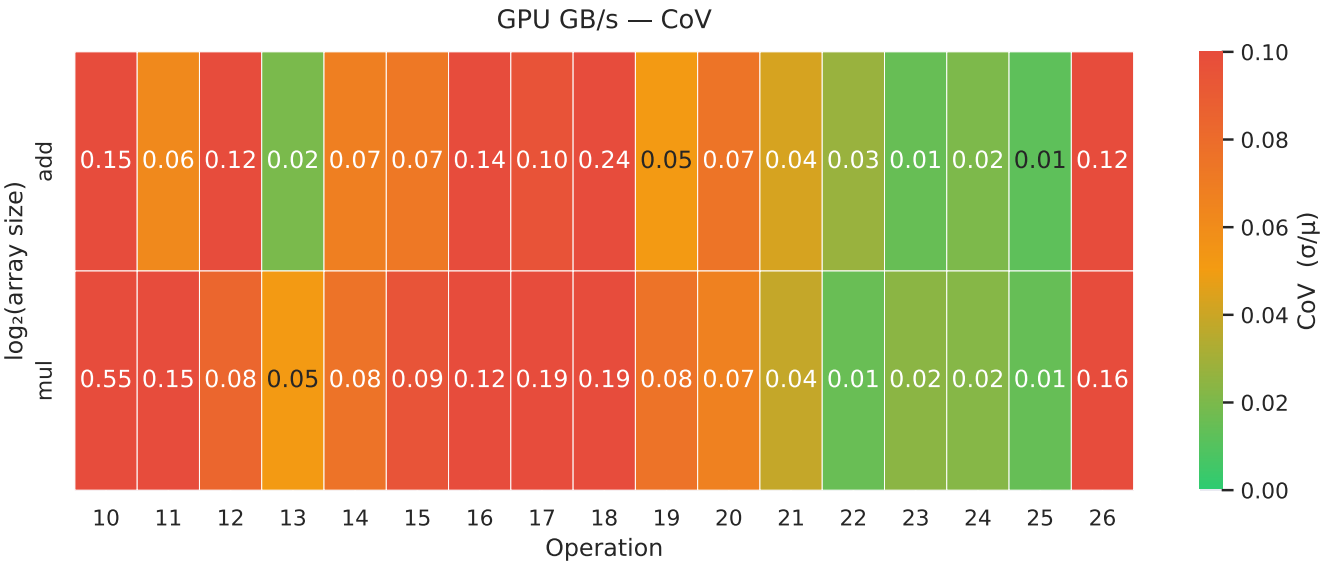


Figure 10:

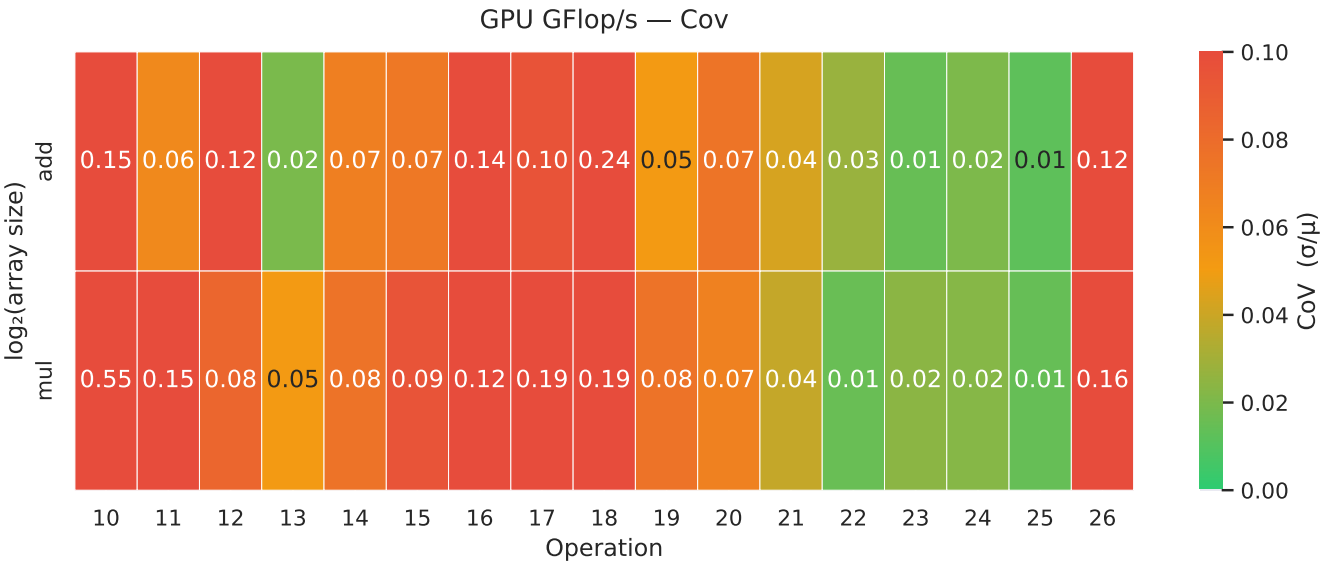


Figure 11:

9.2 Part 2

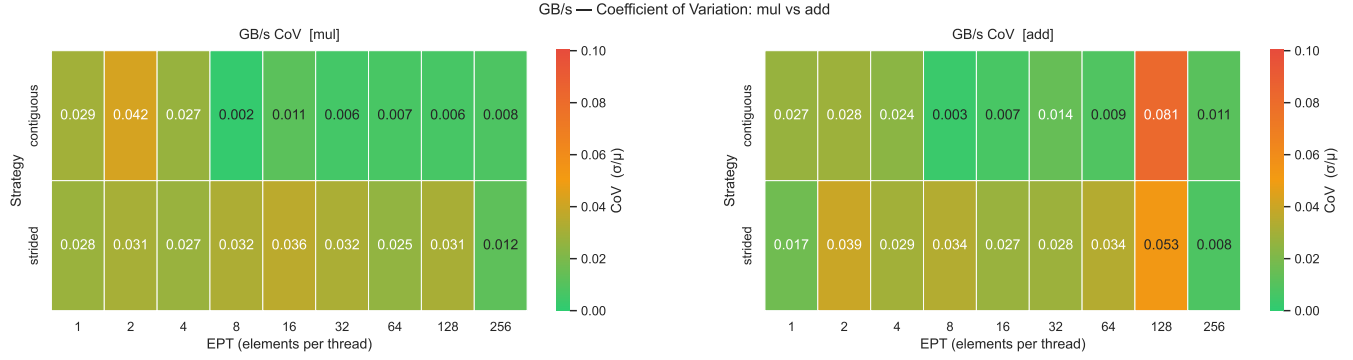


Figure 12:

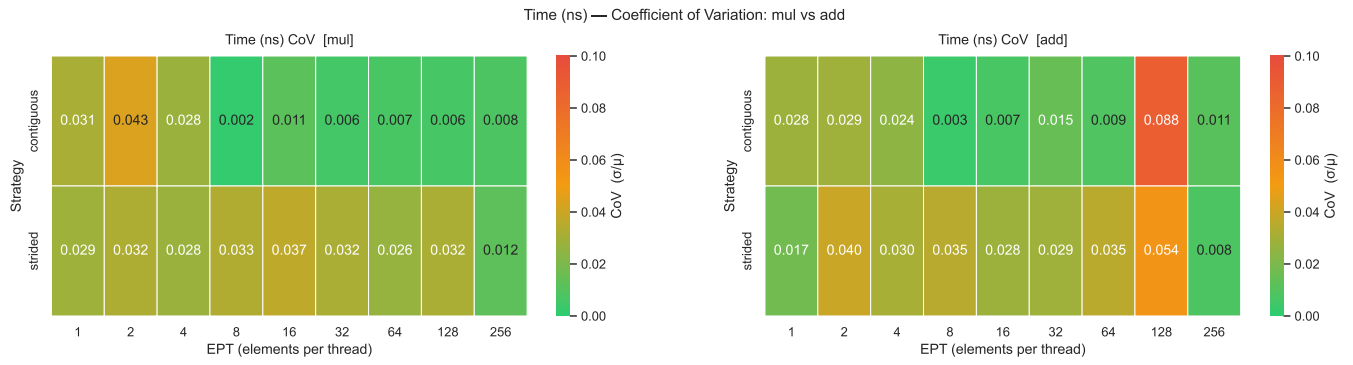


Figure 13:

9.3 Part 3

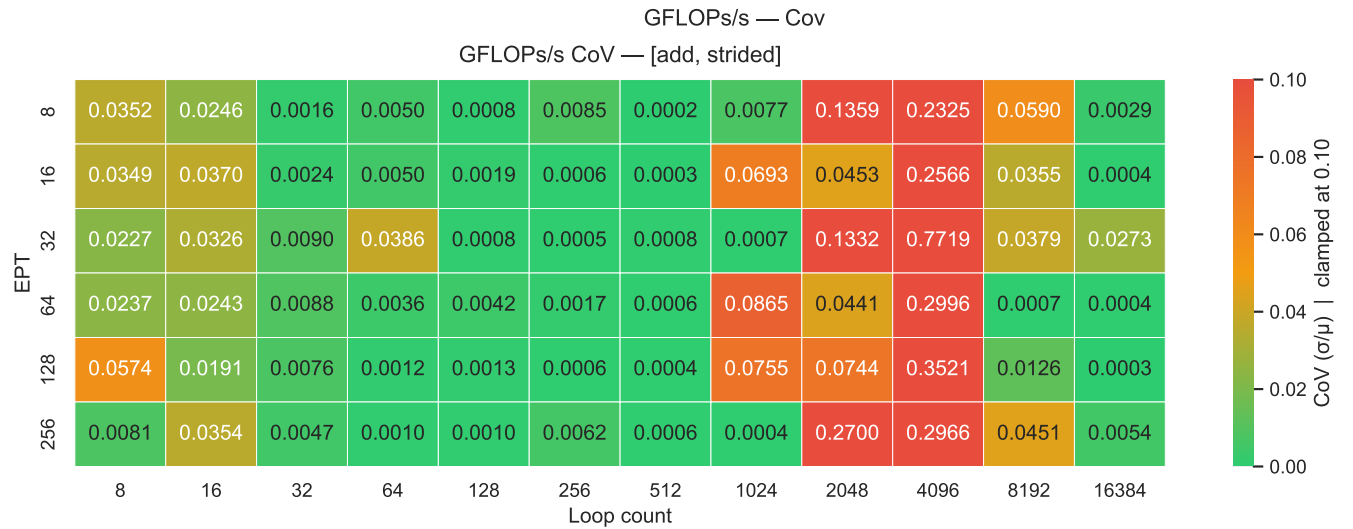


Figure 14:

9.4 Specifications

9.4.1 Macbook

Part	Value
CPU	M2 Pro (6 performance and 4 efficiency)
RAM	16GB
OS	TODO

Table 1: Macbook Specifications

9.4.2 Desktop

Part	Value
CPU	Ryzen 9 5950X
GPU	RTX 3070
OpenCL	300 (Set in code)
Driver Version	591.59
RAM	64GB (3200 Mhz)
OS	Windows 11 - Version 10.0.22631 Build 22631

Table 2: Desktop Specifications

Bibliography

- [1] P. A. Paolillo, “Lecture 2 - Metrics, Distributions & Experimental Rigor - How Many runs do I need?.” p. 38, 2025.
- [2] P. A. Paolillo, “Lecture 2 - Metrics, Distributions & Experimental Rigor - Coefficient of Variation (CoV).” p. 25, 2025.